Benchmark Report

Benchmarking of NQCH’s quantum computer

August 11, 2026

1. Report of Changes

Platform: sinq20
Calibration-id: *de1f87c6abc1f2380dca3875e5c1ed567a482e15*
Calibration date: *2026-08-11 05:11:14*
Calibration note: chore(sinq20): sq recal 110826

Experiment-id: *20260811212203*
Experiment date: *2026-08-11 21:22:04*
Experiment note: *temporary note!!!*

Platform: sinq20
Calibration-id: *4dc4082f38a53222b3956c22202d32a520d4bc78*
Calibration date: *2026-01-15 02:03:03*
Calibration note: chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

Experiment-id: *20260118161538*
Experiment date: *2026-01-18 16:15:38*
Experiment note: *temporary note!!!*

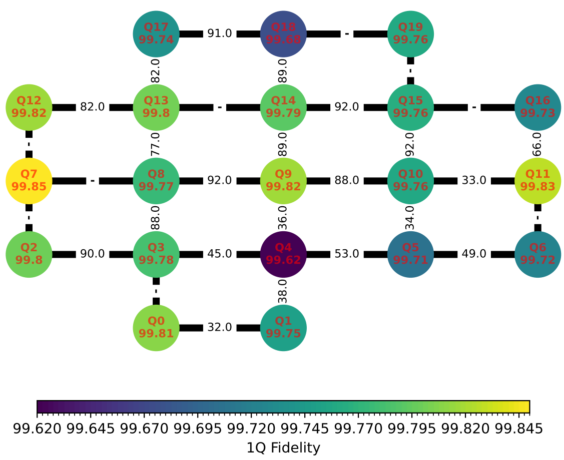
2. Version Comparison

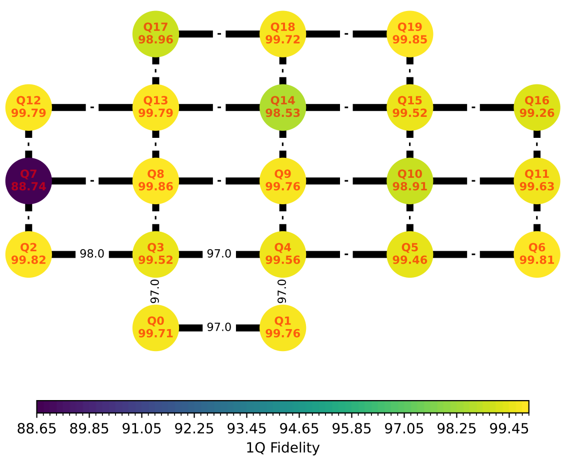
Library	Version	Library	Version
qibo	0.3.3	numpy	2.5.1
qibolab	0.2.17	qibocal	0.2.6
matplotlib	3.11.1	scipy	1.18.0
scikit-learn	1.9.0	pandas	2.3.3
networkx	3.6.1	sympy	1.14.0
torch	2.13.0		

Library	Version	Library	Version
qibo	0.2.22	numpy	2.2.6
qibolab	0.2.9	qibocal	0.2.4
matplotlib	3.10.3	scipy	1.15.3
scikit-learn	1.6.1	pandas	2.2.3
networkx	3.4.2	sympy	1.14.0
torch	2.7.0		

3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".





4. Statistics

	Average	Median	Min	Max
T1 (ns)	-	-	-	-
T2 (ns)	-	-	-	-
Fidelity	0.899	0.998	0.5	1
RO Fidelity	0.5	0.5	0.5	0.5
2q RB Fidelity	-	-	-	-
2q CZ Fidelity	-	-	-	-

	Average	Median	Min	Max
T1 (ns)	1.28e+04	1.23e+04	646	3.65e+04
T2 (ns)	2.36e+25	3.68e+03	125	9.43e+26
Fidelity	0.99	0.997	0.887	0.999
RO Fidelity	0.794	0.777	0.777	0.927
2q RB Fidelity	0.965	0.972	0.923	0.981
2q CZ Fidelity	0.975	0.974	0.967	0.991

5. Best Qubits Selection

k-qubits	Best Qubits	Fidelity
2	10, 15	0.924
3	10, 14, 15	0.920
4	10, 14, 15, 18	0.911
5	10, 14, 15, 17, 18	0.911

k-qubits	Best Qubits	Fidelity
2	2, 3	0.981
3	0, 2, 3	0.976
4	0, 1, 2, 3	0.970
5	0, 1, 2, 3, 4	0.965

6. Summary metrics

Benchmark	Metric	Value	Runtime
Mermin	Max	-	-
Grover 2q	Max prob	0.866	3.12 sec

Benchmark	Metric	Value	Runtime
Mermin	Max	-3.316	1.02 sec
Grover 2q	Max prob	0.89	2.36 sec

7. Randomized Benchmarking Results

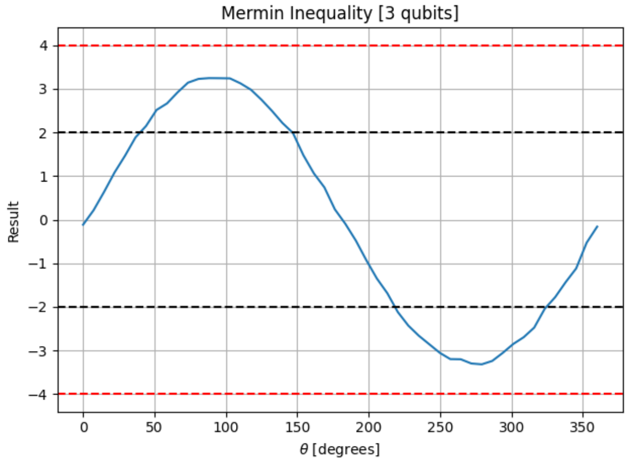
Qubit n	Fidelity	Error Bars	Qubit n	Fidelity	Error Bars
0	0.998	± 8.62e-05	10	0.998	± 0.000103
1	0.997	± 0.000103	11	0.998	± 6.63e-05
2	0.998	± 9.66e-05	12	0.998	± 6.58e-05
3	0.998	± 0.000153	13	0.998	± 0.000102
4	0.996	± 0.000226	14	0.998	± 8.09e-05
5	0.997	± 0.000218	15	0.998	± 0.000103
6	0.997	± 0.000121	16	0.997	± 0.000162
7	0.998	± 4.57e-05	17	0.997	± 8.82e-05
8	0.998	± 0.000107	18	0.997	± 0.000138
9	0.998	± 0.000117	19	0.998	± 8.11e-05

Qubit n	Fidelity	Error Bars	Qubit n	Fidelity	Error Bars
0	0.997	± 0.000131	10	0.989	± 0.000582
1	0.998	± 7.47e-05	11	0.996	± 0.000218
2	0.998	± 7.57e-05	12	0.998	± 0.000126
3	0.995	± 0.000163	13	0.998	± 7.15e-05
4	0.996	± 0.000185	14	0.985	± 0.00183
5	0.995	± 0.000147	15	0.995	± 0.000469
6	0.998	± 8.34e-05	16	0.993	± 0.000558
7	0.887	± 0.0287	17	0.99	± 0.000502
8	0.999	± 4.48e-05	18	0.997	± 0.00026
9	0.998	± 7.2e-05	19	0.998	± 9.05e-05

8. Mermin

Runtime	Qubits	Max
-	-	-

Runtime	Qubits	Max
1.02 sec	0, 2, 3	-3.316



9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in [[10, 15]].

Runtime	Qubits	Max prob
3.12 sec	10, 15	0.866

Runtime	Qubits	Max prob
2.36 sec	2, 3	0.89